Solid State

① Amorphous solid → pseudo solids / super cooled liquid.
Short range order
No definite enthalpy of fusion
No sharp melting point.
Isotropic

② Types of Solids →
(i) molecular – Polar — eg. H_2SO_4
– Non polar — eg. Cl_2, H_2
– H-Bonding — eg. H_2O
(ii) Ionic — eg. NaCl
(iii) Metallic — eg. Cu, Zn
(iv) Covalent / Network — eg. SiC, diamond

③

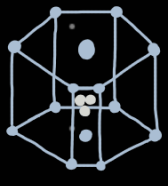
Crystal	Edge	Angle	Lattice
a. Cubic	$a=b=c$	$\alpha=\beta=\gamma=90^\circ$	Primitive, BC., FC.
b. Tetragonal	$a=b\neq c$	$\alpha=\beta=\gamma=90^\circ$	Primitive, BC
c. Orthorhombic	$a\neq b\neq c$	$\alpha=\beta=\gamma=90^\circ$	Primitive, BC, FC, EC.
d. Hexagonal	$a=b\neq c$	$\alpha=\beta=90^\circ$ $\gamma=120^\circ$	Primitive
e. Rhombohedral / Trigonal	$a=b=c$	$\alpha=\beta=\gamma\neq 90^\circ$	Primitive
f. Monoclinic	$a\neq b\neq c$	$\alpha=\gamma=90^\circ$ $\beta\neq 90^\circ$	Primitive, EC.
g. Triclinic	$a\neq b\neq c$	$\alpha\neq\beta\neq\gamma\neq 90^\circ$	Primitive.

↳ Chatorr → all dissymteric!

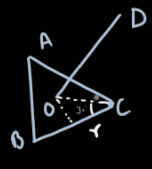
↳ Class Twelth waale 0 ! Hritik Roshan matt bano

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④ Unit cell

Unit cell	Touching Pattern	No. of spheres	C.no.	Packing Eff.
a. Simple Cubic 	edge ($a=2r$)	1	6 2nd nearest → 12	52.4 %
b. Body Centre 	Body diag. ($a\sqrt{3}=4r$)	2	8 2nd nearest → 6	68 %
c. Face Centre (ccp) ABCA 	Face diag ($a\sqrt{2}=4r$)	4	12 2nd nearest → 6	74 %
d. Hexagonal c.p. ABAB 	side main 6, upar 3 neeche 3 ($a=2r$)	6	12 2nd nearest → 6	74 %

↳ * $h = 4\frac{\sqrt{2}}{\sqrt{3}}r$ * $A = 6\sqrt{3}r^2$ * $V = 24\sqrt{2}r^3$



$OD^2 = OC^2 + CD^2 = \frac{8}{3}r^2$ $\therefore OD = \frac{2\sqrt{2}}{\sqrt{3}}r$

$OC = r/\cos 30$, $CD = 2r$

* $P.E = \frac{\rho_{lattice}}{\rho_{sphere}}$

$= \frac{d \cdot N_A \cdot \frac{4}{3}\pi r^3}{MM}$

⑤ Voids

	r_+/r_-
a. linear void	< 0.155
b. Triangular void	0.155 to 0.225
c. Tetrahedral void	0.225 to 0.414
d. Octahedral void	0.414 to 0.732
e. Cubical void	0.732 to 1

In fcc → 4 oct. voids + 8 tetr.

hcp → 6 oct. + 12 tetr.

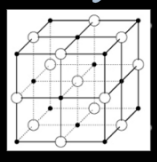
bcc → 6 oct. + 12 tetr. → 0.29 R

↳ .15R

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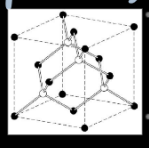
⑥ Ionic Solids

a. NaCl (Rocksalt) → Cl^- in fcc
Na^+ in all oct. voids



(OR, opposite) C.No ↘
Cl^- → 6 Na^+ → 6

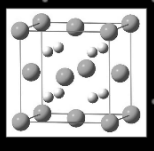
b. ZnS (Sphalerite) → S^{-2} in ccp
Zn^{+2} in alt. tetr. voids



C.N → S^{2-} → 4 Zn^{+2} → 4

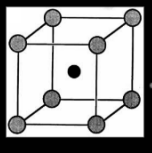
If all 'C' ↙
↳ diamond.
(PE = 34%)

c. CaF_2 (Fluorite) → Ca^{+2} in ccp
F^- in all tetr. voids



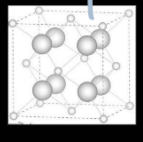
(OR, F^- → simple cubic
Ca^{+2} → alt. cubical voids)
C.No ↘
F^- → 4 Ca^{+2} → 8

d. CsCl (BCC) → Cl^- in simple cubic
Cs^+ in cubic void



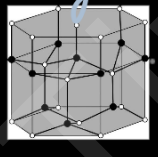
(OR, opposite) C.No ↳ 8:8

e. Na_2O (Anti fluorite) → O^{2-} in fcc
Na^+ in all tetr. voids



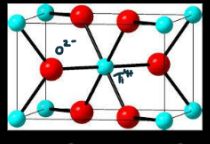
C.No → Na^+ → 4
↳ O^{2-} → 8

f. ZnS (Wurzite) → S^{2-} in HCP
Zn^{+2} in alt. tetr. voids



C.No.
↳ 4:4

g. Rutile structure → TiO_2



C.No → Ti^{4+} → 6 → in oct. fashion
O^{2-} → 3 ↳ Surrounded by 3 Ti^{4+} in △

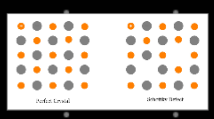
h. Spinel Structure → AB_2O_4
O^{2-} in fcc
A^{2+} in 1/8 tetr. void
B^{3+} in 1/2 Oct. void

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⑦ Defects

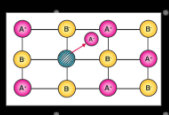
1. Stoichiometric

Schottky



* Similar size
* $\rho \downarrow\downarrow$
* NaCl, KCl, CsCl, KBr. AgBr

Frenkel



* Size gap
* $\varepsilon =$, Dielectric const ↑
* ZnS. AgI, AgCl. AgBr

2. Non-Stiochiometric

Metal Excess

↓

NaCl → yellow
KCl → violet
LiCl → Pink
ZnO → yellow

Metal Deficiency

↓

$Fe^{+2} \rightarrow Fe^{+3}$

3. Impurity

NaCl + $SrCl_2$ ←

Each Sr^{+2} ion causes 1 Cationic vacancy.

⑧ Magnetic Property

a. Paramagnetic → Weakly attr. by $\vec{B}$. eg. NO, O_2, Cr, Mn, Fe, Ni, Co, etc.
(unpaired e^-)

b. Diamagnetic → weakly repelled by $\vec{B}$. eg. Zn, Cd, N_2, NaCl, etc.
(paired e^-)

(I) Ferromagnetic → ↑↑↑↑↑ e^- in presence of $\vec{B}$. eg. Fe, Co, Ni, CrO_2
Strongly attracted, permanently magnetised.

(II) Antiferromagnetic → ↑↑↓↓↑↓ e^- cancel the effect of $\vec{B}$. eg. MnO, Mn_2O_3, MnO_3
No effect.

(III) Ferromagnetic → ↑↓↓ ↑↑ Unequal amount of e^- get aligned.
eg. Fe_3O_4, $MgFe_2O_4$, $ZnFe_2O_4$

→ At high T., all 3 behave like paramagnetic.

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(9) Semiconductors

Doping → Electronic defects

a) p-type → Gr 13 doped in Gr 14. ⇒ conduction via **holes**

b) n-type → Gr 15 doped in Gr 14 ⇒ conduction via **free e^-**

(10) Piezoelectricity

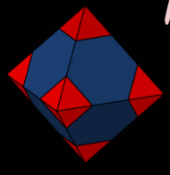
↳ electricity generated due to mechanical stress.

(11) Bragg's Law

↳ $2d \sin\theta = n\lambda$

(12) Truncated shapes

↳



Octahedron



Cube



tetrahedron

(13)

P ↑ ⇒ Coordination no. ↑↑ eg. 6:6 NaCl → 8:8 CsCl structure at high P.

T ↑ ⇒ Coordination no. ↓↓ eg. 8:8 CsCl → 6:6 NaCl at T = 760K

(14) Topmost layer → matlb pure structure mein jo topmost layer ke atom hai.

C.No of an atom in the topmost layer in CCP = 12

↳ 9 ⇒ 3 neeche
6 saath mein

(15) HCP (ABA) CCP (ABC)

HCP ↳ 6 saath wale layer mein
3 upar
3 neeche

CCP ↳ either 3+6+3 or 4+4+4

} ⇒ dono same layers se banti hain



↳ puri layer ko ek ke upar ek
ABAB.... ya phir ABCABC....

(16) Glass is a supercooled liquid.

(17) Rational formula

↳ Na_4Cl_4 not NaCl

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